A checked degree reduction by grouping pairs of bits

DEC-20261029-0857a9 — approve scoped archival acceptance

One admissible fresh blind reviewer independently derived all eight joints. Coordinator comparison supports the frozen producer's **even-width, raw-S3** statement. The original review remains procedurally failed. This is a symbolic result about explicit generators; official archival effect requires the separate native ledger binding.

Accepted scope and inference

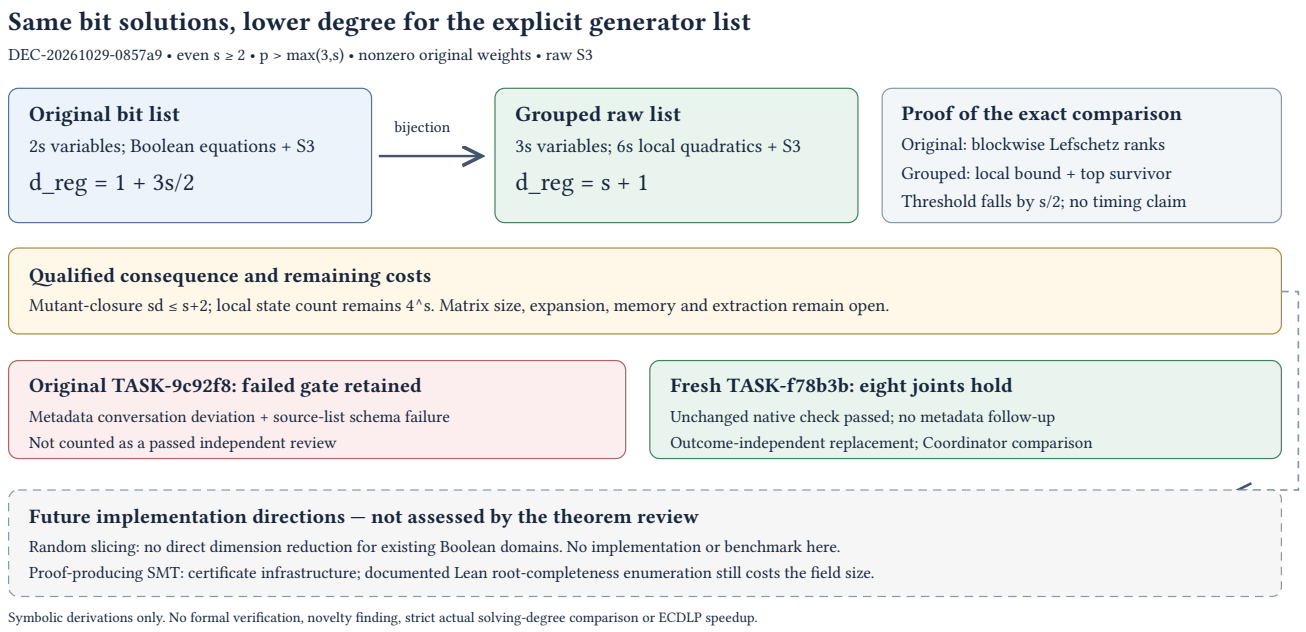
Let p > max(3,s), even s ≥ 2, with nonzero original weights in two disjoint s-bit affine abscissa maps. Retain the nonsingular short Weierstrass curve and finite rational target R=(r,t), t ≠ 0. F0 lists its 2s Boolean equations and raw fixed-target S3. Fg groups pairs within each coordinate, uses three indicators for the nonzero words, lists their Boolean and pairwise-product equations, and substitutes into raw S3. It has 3s variables and 6s local quadratics plus S3; no circuit auxiliaries or affine reduction.

For one pair, zero and the three unit vectors encode the four words. Projection b0=u10+u11, b1=u01+u11 has inverse u10=b0(1−b1), u01=(1−b0)b1, u11=b0b1. Exact bit models and affine quotient rings correspond. Ambient ideals are in different rings; coordinate injectivity is a separate premise.

$$d_{\mathrm{reg}}(F_0) = 1 + \frac{3s}{2}, \quad d_{\mathrm{reg}}(F_g) = s + 1$$

$$d_{\mathrm{reg}}(F_0) - d_{\mathrm{reg}}(F_g) = \frac{s}{2}$$

Degree here is the first component filled by the ideal of the **listed generators' highest homogeneous parts** [1]. The source-qualified mutant-closure corollary is sd_sigma(Fg) ≤ max(s+2,4)=s+2 [3]. The raw quartic input still matters at s=2. This gives no strict ordering of actual solving degrees.



Symbolic proof and custody diagram, not measured performance. Future literature directions are outside the acceptance chain.

Comparison with the frozen producer

The local top quotient is a tensor product of s factors $k \oplus k^3$ with square-zero positive part, giving Hilbert series $(1+3T)^s$. Its top degree survives after quotienting each factor by the corresponding coordinate contribution, which kills the raw quartic top. Together with the local bound this proves the exact grouped value. The original threshold follows separately from weighted blockwise Lefschetz at $p>s$, with exact ranks and a nonzero preceding-degree component [2].

Joint	Verdict	Coordinator comparison
Local models	holds	Zero and unit states over any field; zero word retained.
Bijection/ideals	holds	Explicit inverse and quotient isomorphism; ambient ideals differ.
Counts	holds	Actual widths; 3s variables and 6s local quadratics for even s.
Hilbert series	holds	Local series distinguished from the full highest-part quotient.
Regularity	holds	General full upper bound G+1; exact frozen even-s raw value.
Baseline	holds	Blockwise maximal ranks and nonvanishing prove equality.
Solving degree	holds	Specified mutant-closure upper bound; input degree retained.
Cost/scope	holds	More variables and unchanged state count; no runtime inference.

The general local construction over any field gives $H(T)=\prod_g(1+(2^{w_g}-1)T)$, exact local threshold G+1, and only an upper bound for arbitrary added equations. The seven false-control classes are addressed: missing pairwise products; missing zero word; unanalyzed auxiliaries; missing characteristic/weight premises; constant one; growing width; and unqualified matrix/runtime transfer. For example, adjoining $z=u^2$ to u^2-u leaves highest-part ideal (u^2) in $k[u,z]$, so graded finiteness need not survive a reconstructible auxiliary.

Both reviews additionally derive odd-width and locally reduced-list results, including a separate s=2 proof. Their common formula $2\,\mathrm{ceil}(s/2)+1$ is retained as a **reviewer-derived extension outside this decision's accepted frozen producer scope**. No broader acceptance or promotion is inferred.

Observation, procedure and retained failure

Producer TASK-20261029-e95503 was frozen at `25750271e9` and snapshot-bound at `6a298121de`. Fresh TASK-20261029-f78b3b was sealed at `f282866e49`; its unchanged native checker passed, all input hashes match, and it records no administrative follow-up or deviation. Exact identities and the full receipt are in `grouped-review-inputs.json`.

Original TASK-20261029-9c92f8 failed on its structured source list and its truthful disclosure of a metadata exchange barred by the original plan. Its files and exact failure remain unchanged. No mathematical unblinding is alleged; exact original reply bytes/time are unavailable after compaction. The fresh replacement was authorized at `095e8d3a51` **before root received original verdicts or read the outputs**. All eight questions and controls remained unchanged. Its new prospective administrative exception was not used. The original remains failed under the native failure-provenance archive route.

Count **one admissible blind review**, not two passed reviews. Coordinator comparison is not another independent reviewer or a line-by-line independent audit. Model/effort were requested as native gpt-6.1-sol/xhigh, without provider authentication or an exposed opaque UUID.

Limitations and next actions

The local quotient still has dimension 4^s, equal to the original assignment count. Fixed-width local construction is linear, but expanded S3 can require $O(N^4)$ terms. Growing width incurs exponential indicators and constraints. Elimination columns, rows, fill-in, memory, field arithmetic, reconstruction and checking remain unmeasured. A compact circuit with added variables needs a new graded proof.

No proof-assistant verification, novelty audit, first/last-fall-degree improvement, strict actual solving-degree comparison, solver speedup, unknown-scalar recovery or general prime-field ECDLP advantage is established. Historical PNEXT `invalid_output` and all old statuses remain unchanged. Its empirical outputs are not proof evidence here.

Root next renders and inspects every PDF page, completes actual metadata and verifies the twelve-path ledger archive. Any implementation comparison, wider accepted theorem or formalization needs separate prospective scope and full costs. GOAL-PFDR-fd09e0 remains active; no H/EV/KN promotion is made.

Existing `diagram.svg` and `review-summary.svg` are historical two-summand records. `next-leads.svg` is its earlier candidate snapshot. Their labels were inspected and left unchanged; no shared frontier graph is changed by this bounded TASK decision.

Implementation directions not assessed by this theorem review

Randomized finite-field slicing returns a checked SAT witness or UNKNOWN and targets positive-dimensional systems [4]. Our Boolean domains are already zero-dimensional or empty, so its dimension-reduction mechanism supplies no direct benefit here.

Proof-producing finite-field SMT offers cvc5/FFPACHECK/Carcara and Lean-SMT infrastructure [5]. The documented Lean complete-root check evaluates the whole field; efficient verified GCD remains future work. Neither implementation was installed or run here, and no published performance was reproduced. Benchmark counts are omitted because the author-copy introduction and Table I disagree. The narrow materialized-path search is not a whole-repository novelty or never-tried finding. These directions are outside theorem acceptance.

Primary sources and exact record anchors

[1] Caminata–Gorla, Definition 1.4. arXiv:2112.05579v2.

[2] Phuong–Tran, Theorem 2.7. arXiv:2211.13548.

[3] Salizzoni, Definition 2.5 and Proposition 3.10. arXiv:2304.13485.

[4] Goharshady, Hader, Kovacs and Motwani, Randomized Satisfiability Checking for Non-Linear Arithmetic over Finite Fields, Algorithm 1 and Section IV, FMCAD 2026. TU Wien primary record and full text.

[5] Saccomani et al., Proof Production for Satisfiability Modulo Finite Fields with Proof Checking in Pacheck and Lean, Sections III–V and footnote 1, FMCAD 2026. Author PDF. PDF hashes are unknown, not invented.

Producer: `grouped-proof.md`, Sections 2–6. Admissible review: `TASK-20261029-f78b3b/blind-review.md`, Joints 1–8. Failed original: `TASK-20261029-9c92f8/`. Technical records and successor plan: `coordination/prime_sat_regularity_20261029/grouped-review/replacement-01/`. Complete hashes and native check output: `grouped-review-inputs.json`.